4 **(a)** Three identical springs A, B and C of negligible mass are shown in Fig. 4.1. Each spring has spring constant *k*.

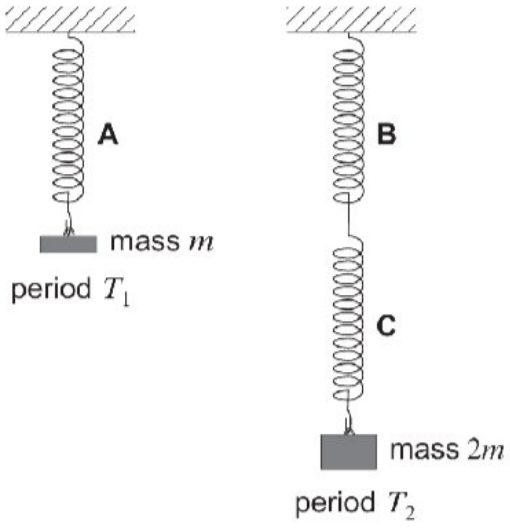


Fig. 4.1

The oscillation of the spring-mass system obeys the simple harmonic equation

$$a = -\left(\frac{k}{m}\right)x$$

where *m* is the mass of the load, *a* is the acceleration and *x* is the displacement from equilibrium position.

For the system on the left, the period of natural oscillations is T_1.
For the system on the right, the period is T_2.

Calculate the ratio of the periods $\frac{T_2}{T_1}$.

ratio = ... [2]

(b) In a separate experiment a mass hanging from a spring is displaced upwards by 30 mm from its equilibrium position and released at time $t = 0$. It performs simple harmonic motion with a period of 1.20 s.

(i) Show that the maximum speed of the oscillating mass is 0.16 m s^{-1}.

[1]

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(ii) **Taking upwards direction as positive**, sketch on Fig. 4.2 the variation with time of velocity for two cycles of oscillations of the mass.

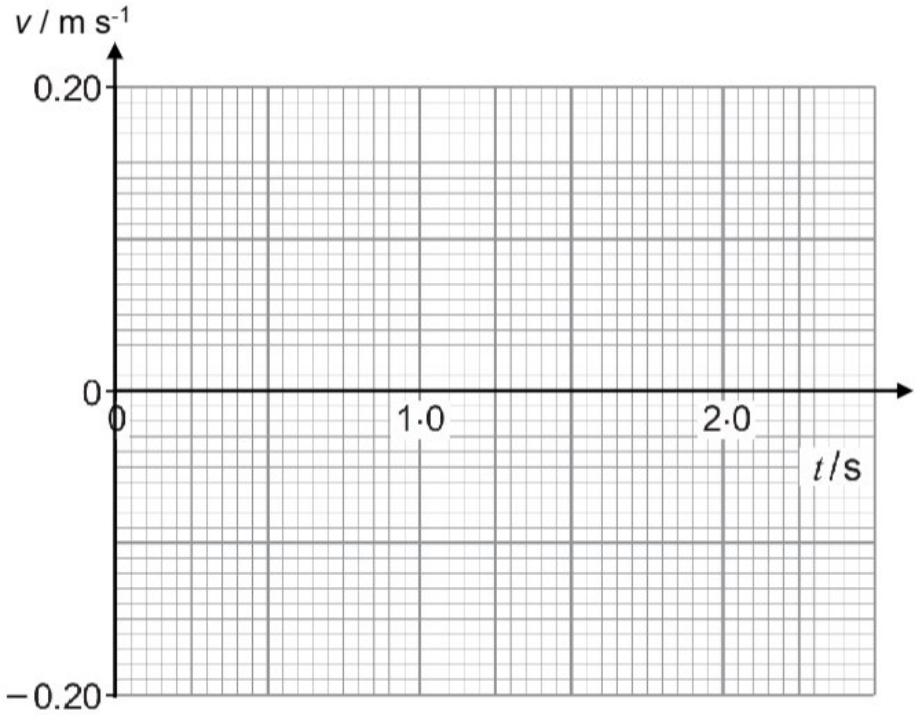


Fig. 4.2

[3]

(iii) Calculate the speed of the mass at $t = 3.50$ s.

speed v = ... m s^{-1} [2]

(iv) A student thinks that the kinetic energy of the mass varies at a frequency of 1.67 Hz. Evaluate this claim.

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..... [2]

[Total: 10]

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